

FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestone 3 Report

**AI-Based Student Dropout Early Warning System for Higher Education:
Reproduction and Cross-Dataset Generalization on UCI and OULAD**

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Submission: Milestone 3 Data Collection and Initial EDA

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1. Updated Executive Summary

The goal of this milestone has been to collect the two project datasets and complete preliminary exploratory data analysis prior to baseline model development. As part of this, the finalization of the UCI Predict Student's Dropout and Academic Success dataset occurred. The reproducing and benchmarking dataset is comprised of 4,424 records and 37 features with no missing values, no duplicate row entries. Following this process, OULAD was gathered as the primary evaluation dataset which includes 7 related csv files that include approximately 32,593 student records; 173,912 assessment records; and more than 10.6 million raw Virtual Learning Environment (VLE) interactions. From an initial exploratory data analysis of both datasets, there appears to be a similar trend. For example, in the UCI dataset, there are several factors significantly related to dropout. Specifically, a higher number of approved curricular units and better semester grade averages are associated with a greater likelihood of success. On the other hand, from the exploratory data analysis in OULAD, it was found that students who withdrew had less assessment participation, lower average test scores, and substantially less VLE engagement compared to either successful students (i.e., those receiving a passing grade) or students who received a distinction. In addition to demonstrating a similar trend between the two datasets, the quality of each dataset was reviewed. Based on this review, several major issues were identified in preparing the OULAD dataset for the next milestone (preprocessing, feature extraction, and baseline modeling). These included: structural missingness within the assessment table in OULAD; multi-table integration across various tables within OULAD; and leakage control within the virtual learning environment (VLE) data in OULAD. Collectively, these findings confirm that the project can now proceed with Milestone 4.

2. Data Collection and Dataset Description

2.1 Finalized Dataset Sources and Acquisition Procedures

Two public datasets were collected for this capstone. The first is the UCI Predict Students' Dropout and Academic Success dataset, which will be used as both the reproduction and benchmark dataset for the base paper by Islam et al. (2025) [1]-[3]. The second is OULAD, which will be used as the primary evaluation dataset since it contains richer behavioral and temporal data to allow for the study of model transferability and robustness [4]. The UCI dataset was retrieved from a single CSV download, while OULAD was retrieved via an official ZIP archive and subsequently unzipped into seven CSV files. Both datasets are stored within a structured project directory to ensure consistent reproducibility of all subsequent preprocessing and modeling activities.

No API downloads were needed to complete this activity milestone. The first verification task was to verify each dataset's ability to read from disk, count rows, identify key table schema attributes, identify targeted variable(s), and detect missing value(s) or duplicate entries in their

respective main tables. No integration was performed on the UCI dataset during collection. Integration was performed on the OULAD dataset with respect to inspecting its relationship-based table structure and preparing for future join operations through use of `code_module`, `code_presentation`, and `id_student`.

2.2 Dataset Structure and Characteristics

Dataset	Role	Format	Observations / Tables	Target Variable	Primary Data Type
UCI Predict Students' Dropout and Academic Success	Reproduction and benchmark dataset	Single CSV	4,424 records / 37 columns	Target: Graduate, Dropout, Enrolled	Structured tabular data
OULAD	Primary evaluation dataset	ZIP with 7 CSV files	32,593 student records; 173,912 assessment rows; 10,655,280 raw Virtual Learning Environment event rows	final_result: Pass, Withdrawn, Fail, Distinction	Structured multi-table learning analytics data

The UCI dataset is very compact and relatively easy to be loaded into models. However, many of the datasets in the UCI dataset that contain numeric information have been encoded as either categorical (nominal) or ordinal variable types. In contrast, OULAD represents a much more complex representation of how a real-world Learning Analytics environment would look like. It contains all sorts of demographic information regarding students, their registration histories, their assessment records and detailed record of the interactions they made with the Virtual Learning Environment during the 2013B, 2013J, 2014B and 2014J module-presentations.

2.3 Variable Description and Data Integration Logic

For UCI, the primary predictor variable families were demographic/background data, academic preparation data, institutional/financial data, and first/second semester academic performance. Based on the UCI dataset, Semester Approved CU, Semester Grades, Admission Grade, Prior Qualification Grade, Age at Enrollment, Debtor Status, Tuition Payment Status and Scholarship Status were identified as the top candidates to use in subsequent modeling.

The critical variables for OULAD spanned multiple tables within the database. The `studentInfo` table contained all key student and target information. The `studentRegistration` table included all enrollment and withdrawal timing fields. The `studentAssessments` table captured all assessment submissions and assessment scores. The `StudentVLE` table recorded VLE behavior at a detailed

level by each student. In order to model this raw data, Milestone 4 required aggregating assessment and VLE behavior into student-level features. All three datasets were linked using code_module, code_presentation and id_student.

OULAD Table	Rows	Planned Use in Project
studentInfo	32,593	Core student attributes and final result target
studentRegistration	32,593	Registration and withdrawal timing features
studentAssessment	173,912	Assessment participation and performance aggregates
studentVle	10,655,280	Student Virtual Learning Environment engagement features after aggregation
assessments	206	Assessment metadata for feature enrichment
courses	22	Course/module reference
vle	6,364	Virtual Learning Environment activity metadata if needed for interpretation

The biggest limitations in data size are not access but analytical structure. Data can be easily interpreted and modeled from UCI, however some of the most powerful predictors in UCI will likely occur too late in time for an actual "early" intervention program. Conversely, OULAD has stronger behavioral signals much earlier on than UCI does; however OULAD would require a larger number of joins, additional levels of aggregation, and additional mechanisms to prevent leakage.

3. Exploratory Data Analysis (EDA)

3.1 Statistical Summary and Initial EDA - UCI Dataset

The UCI dataset contains 4424 rows and 37 columns; of those, 36 represent predictor variables and the remaining column represents the multiclass target variable. All record-based validation indicated there were no missing values, and all row-based validation indicated there were no duplicated rows. Class distribution in the dataset was slightly unbalanced: 49.93% of records belonged to Graduate (2209); 32.12% belonged to Dropout (1421); and 17.95% belonged to Enrolled (794). Thus, the class representing Enrolled will be considered the most sparse in relation to the other two classes when performing model development and evaluations.

Following the initial class-distribution assessments, comparisons of the overall characteristics for each of the groups were made. On average, students classified within the Graduate group received approximately 6.23 units per semester as approval in their first semester and 6.18 units per semester as approval in their second semester; in comparison, students classified within the

Dropout group received on average 2.55 units per semester as approval in their first semester and 1.94 units per semester as approval in their second semester. Additionally, graduate students have consistently exhibited higher semester grades and have a significantly higher rate of being scholarship-holders relative to students classified as Dropouts who exhibit higher debt-to-income ratios and lower levels of payment compliance regarding tuition payments. Collectively, these results suggest that both academic success and financial status can serve as key indicators to predict graduation outcome based upon the given benchmark data.

Results from correlation analysis revealed the same general trend; specifically, the highest correlations with a binary dropout indicator included: Curricular units 2nd sem (grade) (-0.572), Curricular units 2nd sem (approved) (-0.57), Curricular units 1st sem (grade) (-0.481), Curricular units 1st sem (approved) (-0.48), and Age at enrollment (0.254). Therefore, it appears that academic performance through time (i.e., unit completion) and age-related factors associated with enrollment patterns may influence an individual's likelihood to drop-out of college more so than macro-economic variables such as GDP growth rates, inflation rates, or unemployment rates.

Using the IQR rule to identify outliers resulted in large counts of potential anomalies in the age and grade-related variables; including 441 observations in Age at enrollment and over 700 observations in each of the semester-grade variables. Since we are treating these values as potentially valid representations of variability in how different types of students complete coursework, enroll in courses, and perform academically instead of automatically flagging them as erroneous due to extreme variation, we treat these counts as possible legitimate sources of heterogeneity among students.

3.2 Statistical Summary and Initial EDA - OULAD

OULAD is structurally richer than UCI, so this dataset was analyzed at two levels of granularity; first, by examining each table individually and secondly, after merging selected feature-sets together. As previously stated, there are 32593 student records within the central studentInfo table along with twelve variables. The overall distribution of the target variable shows that there are approximately 12361 Pass (37.93%), 10156 Withdrawn (31.16%), 7052 Fail (21.64%) and 3024 Distinction (9.28%) records respectively. The large size of the Withdrawal class confirms the suitability of OULAD for investigating causes of attrition.

Data Quality Inspection revealed no duplicate records within studentInfo, studentRegistration and studentAssessment. However, an important pattern of missing values was observed. There were 1111 (3.41%) missing values in the imd_band field of studentInfo. A further six fields exhibited significant missing value patterns. The date_registration field of studentRegistration had forty-five (0.14 %) missing values, as did the date_unregistration field (which had 22521 (69.10 %) missing values). Most of these missing values arose due to structure of the data set since many students have never withdrawn from their courses. The score field of studentAssessment had seventy-three (0.05 %) missing values. While these results would

certainly impact subsequent pre-processing steps, they do not appear serious enough to render the data set completely unusable.

Following the merge of the primary OULAD tables at a single student record level, some early warning signs regarding potential academic problems became apparent. For example, the average number of assessment records for Distinction students was eight point seven-one (8.71), their average scores per assessment were eighty-eight point four-nine (88.49), and their average total Virtual Learning Environment click counts were twenty-six sixty-six-point-seven-six (2666.76). Conversely, students who withdrew averaged only one point two-nine (1.29) assessment records, averaged scores of twenty-nine point five-zero (29.50) per assessment and averaged only three hundred thirteen point nine-five (313.95) total Virtual Learning Environment click counts. The Pass students were somewhat closer to Distinction in terms of their engagement metrics and assessment averages. An additional observation was that the average amount of studied credit points for students who withdrew was ninety-one point four-three (91.43) which is greater than all other groups examined. Therefore, it appears possible that withdrawal risk could be related to not only low academic engagement but potentially also to high amounts of credit points taken at once or difficult combinations of courses.

As mentioned earlier, the studentVle table required summarization prior to being usable as input to any analytical models. Following aggregation of the event-based Virtual Learning Environment usage records into a model-ready student based behavior metric summary table containing thirty-two thousand-two-hundred-twenty-eight (32228) grouped student records indicated that OULAD can be computationally feasible for local processing using hardware capable of supporting moderate-sized log files and/or if the larger log files can be segmented into smaller portions, processed separately, and then summarized into modeling-friendly formats.

3.3 Data Visualization and Interpretation

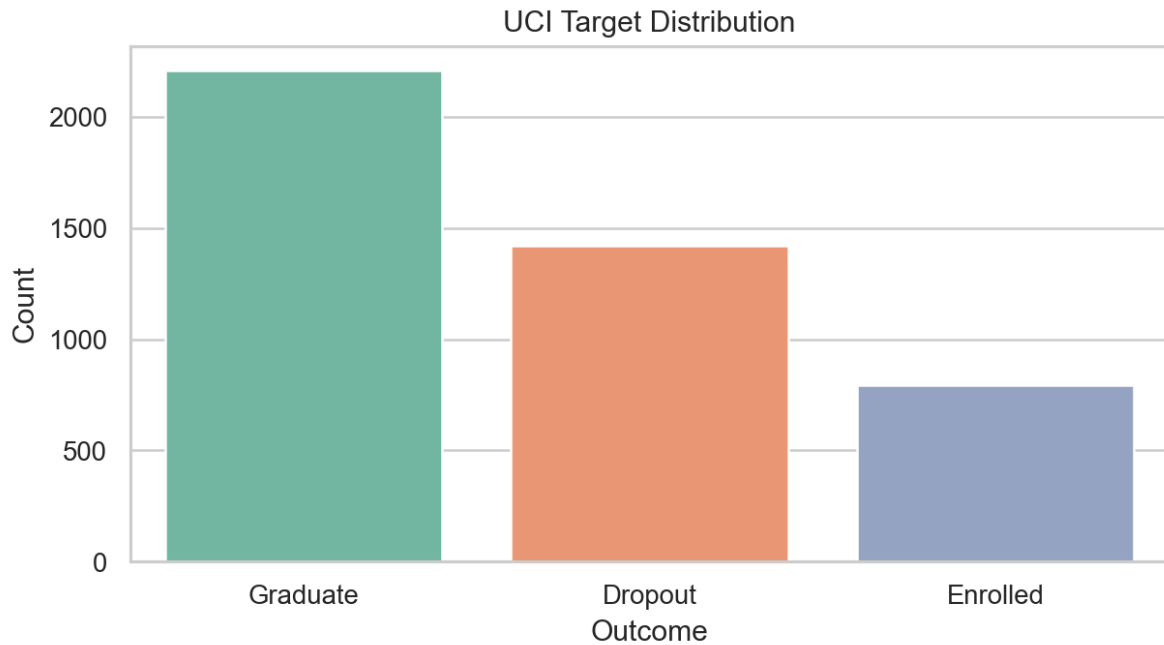


Figure 1. UCI target distribution shows moderate class imbalance, with Enrolled as the smallest class.

Figure 1 demonstrates that the UCI target variable does not present severe imbalance in terms of distribution; however, since there are fewer enrolled students in the Enrolled group compared to Dropouts (in the Dropouts group), this will likely contribute to multiclass classification being significantly harder than a binary classification for the outcome of attrition.

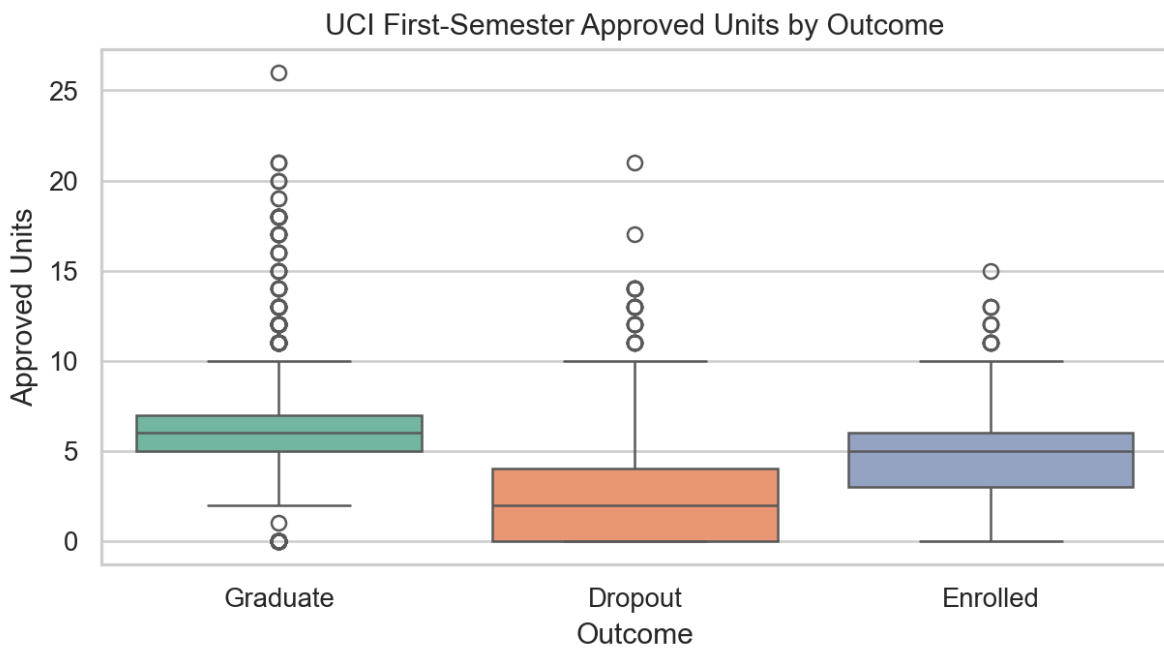


Figure 2. UCI first-semester approved curricular units differ sharply across outcomes.

The results shown in Figure 2 show much clearer separation between Dropout and Non-Dropout classes than was possible with a mean comparison alone. The evidence presented in Figure 2 indicates that lower amounts of approved curricular units are highly correlated with the Dropout class. Therefore, we can determine that approved curricular units should be one of our benchmark features.

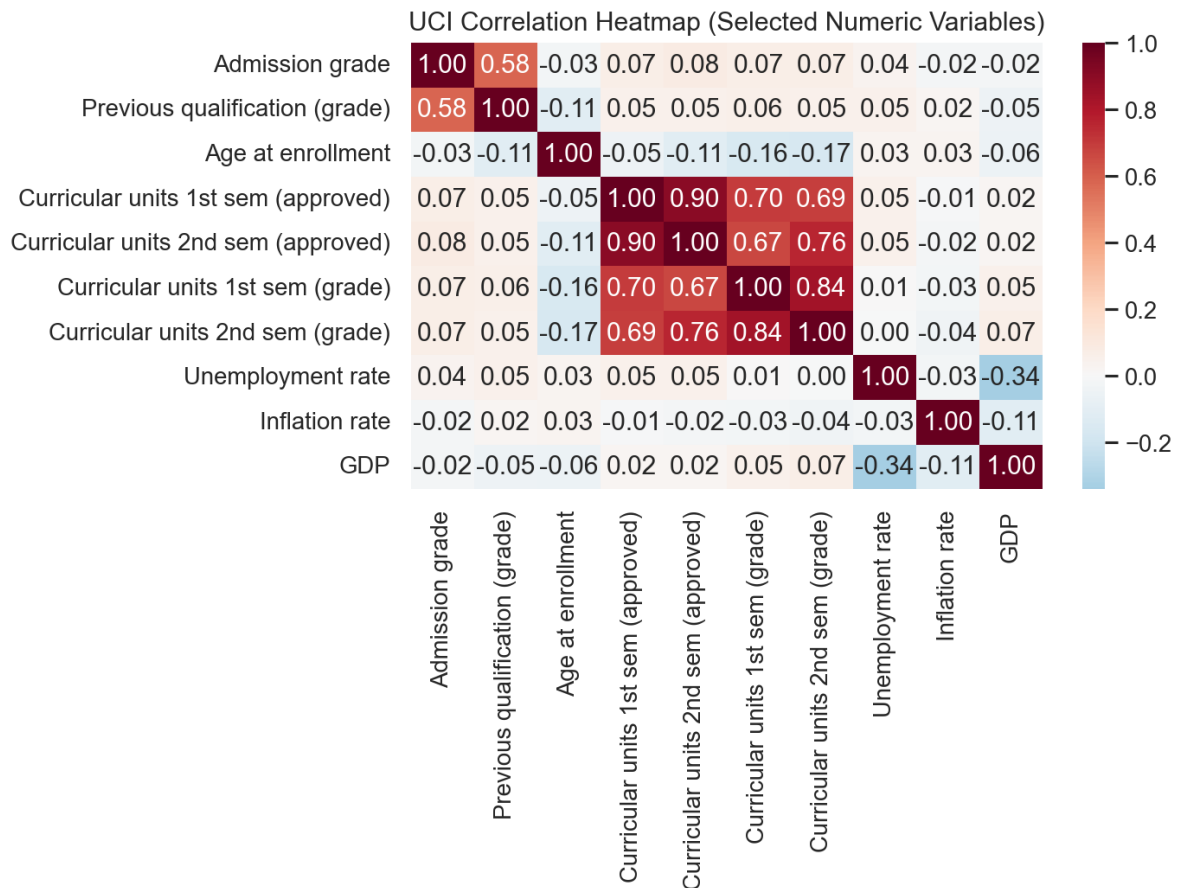


Figure 3. UCI selected-variable heatmap shows strong positive relationships among semester performance variables.

The results shown in the heat map show many of the academic indicators are highly correlated with each other; especially when looking at approved units and grades across both semesters. While this will be helpful for predictive modeling, this correlation will require more work on the part of baseline modelers (feature selection) as well as consideration for multicollinearity.

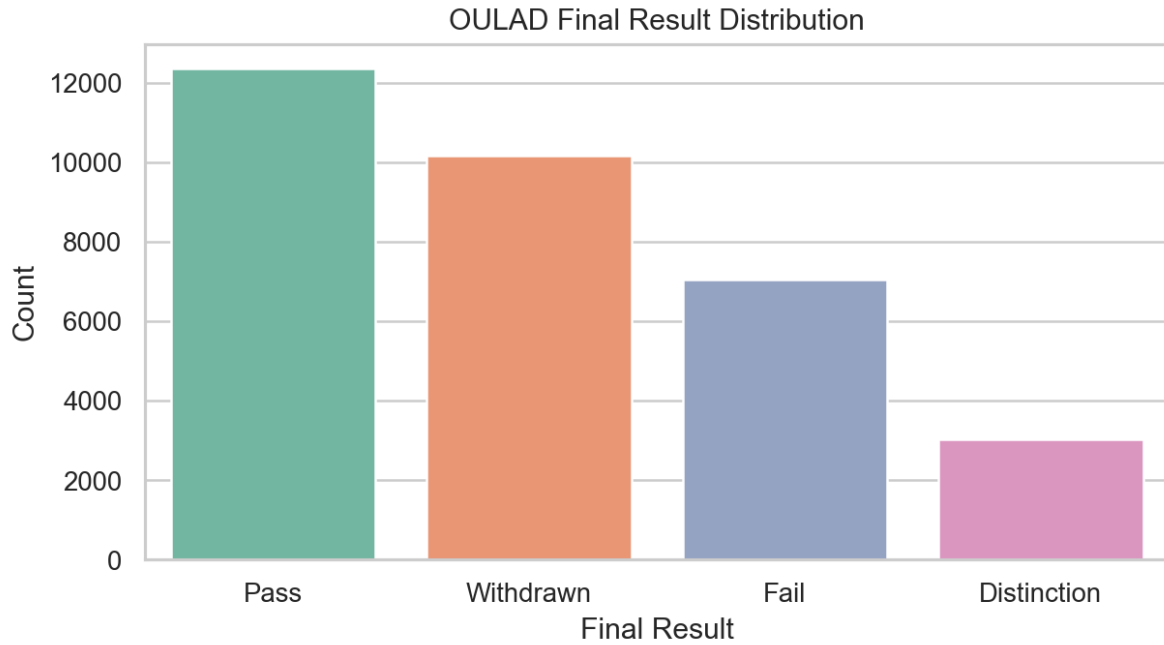


Figure 4. OULAD final_result distribution supports a strong attrition-oriented evaluation setting.

The Figure (4) illustrates one of the primary advantages of using OULAD for this study. There are sufficient numbers of students who withdraw from the course to allow for meaningful early warning systems. In addition to a simple "Withdrawal" class there are three classes for outcomes (Pass, Fail, Distinction). These three classes provide a greater amount of variation among outcome classes than would a single binary class.

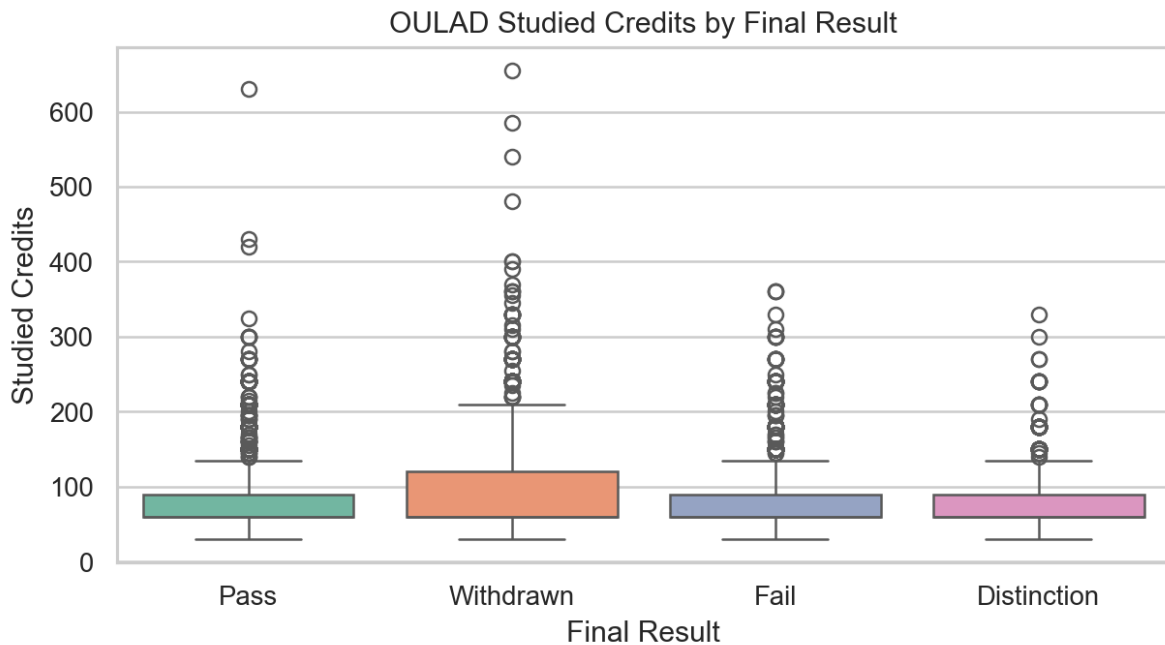


Figure 5. OULAD studied credits vary across final outcome groups, with the Withdrawn group carrying the highest average load.

This visualizes that Studied_Credits may be an appropriate contextual factor, rather than an indicator of risk itself. The fact that a high number of credits at study time indicates a greater likelihood of withdrawal (or failure) does not suggest that students with heavy credit loads are generally failing. Instead, it seems likely that they are experiencing more pressure and possibly have more tenuous student workload patterns.

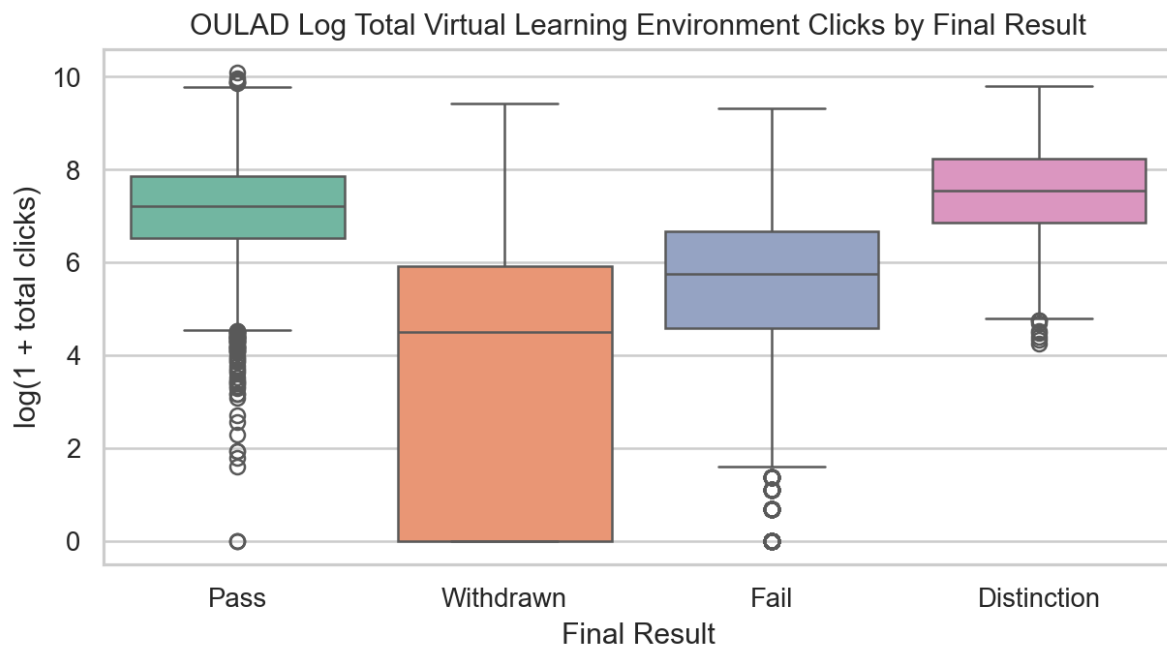


Figure 6. OULAD log total Virtual Learning Environment clicks show that withdrawn students are substantially less active online.

The finding in Figure 6 is one of the most clear-cut from the Milestone. There is significantly less virtual learning environment (VLE) usage by the withdrawn group compared to both pass and distinction. Thus, we can draw upon the idea of behavioral engagement early in the course as a central characteristic family for building predictive models related to later outcomes.

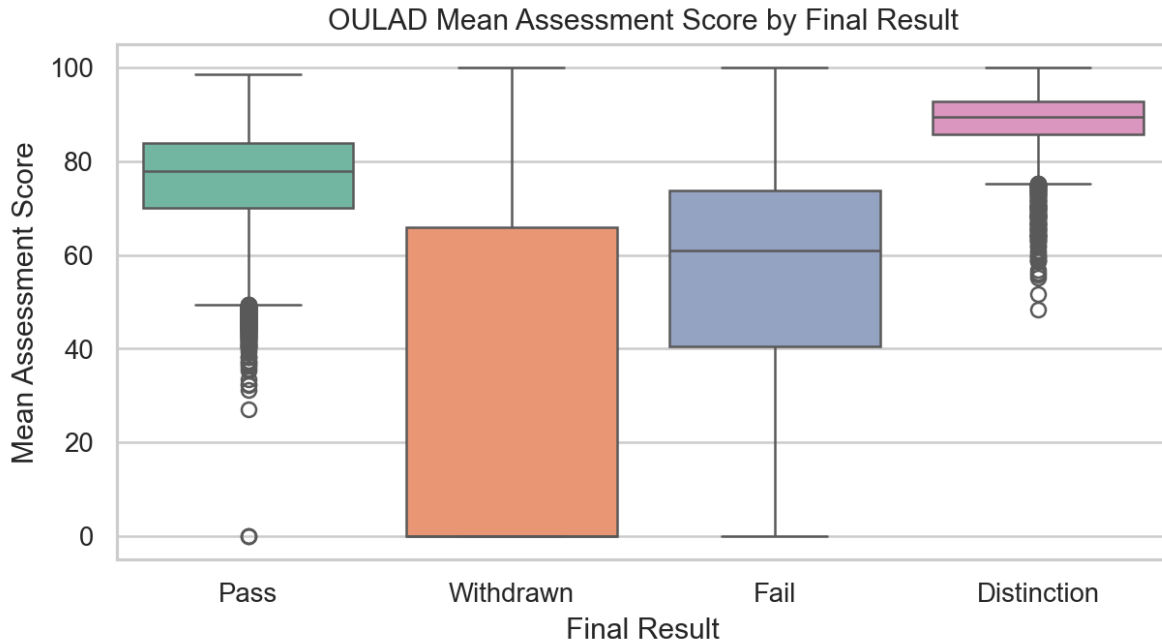


Figure 7. OULAD mean assessment score also separates Withdrawn students from the more successful groups.

In addition, the data presented in Figure 7 demonstrates that there appears to be some level of mutual support between academic success and behavioral engagement in OULAD. Specifically, students who will ultimately drop out of courses not only logged into their VLEs less frequently but also submitted fewer assignments and achieved lower scores on those they did submit.

3.4 Initial Business Insights and Analytical Interpretation

Early insights about business were made through this study. First, both of these studies provide evidence in support of one key business issue: Identify at risk students prior to their withdrawal as final. Fields of Financial Progress and Academic Progress are especially indicative of student retention within the UCI dataset. Assessment Participation, Assessment Performance and Behavioural Engagement appear most indicative of at-risk status within the OULAD dataset. Therefore, even with identical objectives (business), institutions utilizing different data environments will likely have different feature strategy requirements.

Secondly, the analysis has provided evidence of a critical practical distinction between Benchmark Accuracy and Intervention Timing. The UCI dataset contains very strong second semester performance field variables; however, those variables may be too late to enable truly early action. Conversely, OULAD offers Behaviour Streams that could potentially be cut-off at earlier times allowing it to be used as a more viable early warning system development environment.

Finally, the observed difference in behaviour/assessment between Withdrawn Students and the other Groups within the OULAD dataset supports the use of a common Attrition Outcome in future cross-dataset experiments comparing Persistence vs. Attrition.

4. Data Quality and Preprocessing Assessment

4.1 Data Quality Concerns

Both datasets are suitable for use, however there are significant differences regarding preparation effort. UCI can be used directly from the baseline model since it includes no missing entries, no duplicate records and a simple table structure. The major difference with respect to quality is that many numeric coded entries should have been treated categorically or ordinally rather than continuously. Furthermore, some of the stronger academically based variables could represent time constraints if the objective is true early intervention versus simply replicating benchmarks. The OULAD dataset represents additional practical data quality issues. Although the missing values in the fields `imd_band` and `date_registration` are limited and therefore easily managed (and similarly the missing value in the field `date_unregistration`), the missing values in this last-named field are structural and thus cannot be treated as standard random missing data. There are also a few missing values in the `studentAssessment` score field and while there would not necessarily be a reason to exclude these, the very large size of the `studentVle` table creates numerous scalability, aggregation and consistency problems. Again, these are not reasons to eliminate the dataset as they provide the rationale for why it should serve as your primary evaluation dataset.

4.2 Preliminary Preprocessing Plan

- Reproduce the complete set of features for the baseline paper (with all features) on UCI. Identify the semester-specific features that are late semester variables while discussing the realism of early warning systems.
- Treat all code-based fields appropriately as categorical/ordinal in UCI. Use scaling sparingly on linear baselines (e.g., logistic regression).
- Merge `studentInfo`, `studentRegistration`, `studentAssessment`, and aggregated `studentVle` records at the student-module-presentation level for OULAD.
- Impute or address missing `imd_band` values. Review the few missing scores. Exclude `date_unregistration` from input data for prediction to avoid leakage.
- Define student behaviour features based upon aggregated virtual learning environment activity within a "chunked" window. Define time-sensitive limits for "early warning" feature windows.
- Use attrition vs. persistence labels across datasets for future cross-dataset analyses. Instead of using raw column names to align features, use the family of shared features.

4.3 Anticipated Transformation and Feature Engineering Challenges

The biggest challenge to pre-processing is leakage. In UCI, the last academic variables will typically inflate the model's prediction power for a time while decreasing its ability to predict future actions. Similarly, in OULAD (`date_unregister`), and behaviors that occur late in the

course can also provide an unintended "hint" about what would happen with those students if they were used in the model before some type of time cut-off was applied. The second main issue is how to get all features into one format, which is necessary for modeling across both datasets; there is no common schema between the two, so this will be accomplished through aligning similar concepts including demographic, prior preparation, academic progress, etc., along with early student engagement. A third challenge is scale. As mentioned earlier, OULAD is reasonably easy to compute using available computer resources when it is broken down into smaller pieces or chunks. However, even after breaking it down into chunks, OULAD still has to be aggregated or reduced to prepare it for modeling. Therefore, aggregation and reduction are required steps of pre- processing rather than optional enhancements.

5. Modeling Readiness Assessment

In general terms, the data readiness assessment has been completed successfully. From a baseline reproduction modelling standpoint, UCI is prepared for use at Milestone 4, given typical pre-processing (train/test split, treatment of missing values, selection of multi-class evaluation criteria) are considered. OULAD is also ready to model using, but it is not currently ready as an individual record level ready source of data due to the need to aggregate studentAssessment and studentVLE records into a Student-Level Virtual Learning Environment (VLE), and then filter out potential record-level leakage.

With regard to the availability of resources for processing/computational aspects, the project should remain feasible to perform within typical resource limits available with most laptops or Google Colab environments. Although the largest original table (studentVLE) was processed, the results from this milestone confirmed that chunking-based aggregation would be sufficient to reduce this large table into meaningful behavioral summaries. However, there are two primary overfitting concerns regarding these datasets: UCI's strong later semester variables; and the ability to produce multiple engineered features based upon behavior logs in OULAD. While both could be addressed by careful feature engineering, comparative baseline analysis, and explicit validation processes, they do represent the two greatest methodological risks associated with moving forward to Milestone 4. Therefore, the major risks entering Milestone 4 are primarily related to methodology, including identifying reasonable early warning time frames, minimizing label leakage, managing class imbalance and developing a cross-dataset representation which will support meaningful portability testing in future milestones.

6. References

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